

Dagangan: Delivering opportunity to Indonesia's rural economy at scale

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ABOUT DAGANGAN

Dagangan expanded its hubs across Java and more than 20,000 and forged an AI grow 10-fold in tv through regional with Google Cloud Google Workspace



About Dagangan

Dagangan is an Indonesian rural commerce platform that provides access to daily necessities at affordable prices to rural communities. Founded in 2019 by a group of friends, Dagangan's mission is to accelerate Indonesia's economic journey by empowering communities and providing opportunities for all. The platform focuses on maximizing the potential of remote communities through a tech-powered hub-and-spoke network that enables a nimbler supply chain of high-quality and affordable daily goods. Dagangan prides itself

Google Cloud results

- Supports community empowerment by boosting crash-free user experiences to more than 99% from 92%
- Enables rural expansion mission by powering rise in active daily users from 3,000 to 5,000
- Brings scalable data insight that enables half of small business customers to grow revenue by 60%

Indonesia is a nation on a dynamic prosperity, with an average growth of 5% per year

One-third
in platform
traffic in
1 year

on empowering women entrepreneurs in local communities, while leveraging community influencers to solve access issues, with a commitment to delivery in under 24 hours.

Industries: Professional Services

Location: Indonesia

SQL (<https://cloud.google.com/sql/>).



About Devoteam

Devoteam is a leading IT consulting firm focused on digital strategy, platform technologies, cybersecurity, and business transformation. With more than 25 years of passion for tech and offices in 18 countries across EMEA and beyond, Devoteam believes that technology with strong human values can actively drive change for the better.

Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

(<https://tradingeconomics.com/indonesia/annual-gdp>)

that makes it one of the world's fastest growing economies. Amid this inspiring growth, the economic and digital divide between urban and rural regions is expanding fast as well. For many people in Indonesia's agricultural heartland, access to mega-cities

(<https://knoema.com/atlas/Indonesia/population#:~:text=Indonesia%20%20population%20as%20a%20share%20of%20total%20population%20in%202022%2C%20urban%20population%20annual%20rate%20of%202.39%25.>)

such as Jakarta in search of opportunities.

Google Workspace

(<https://workspace.google.com/>)

Docs

(<https://workspace.google.com/products/docs/>)

Google Drive

(<https://workspace.google.com/products/drive/>)

Google Meet

(<https://workspace.google.com/products/meet/>)

Sheets

(<https://workspace.google.com/products/sheets/>)

In towns and villages across the sprawling Indonesian archipelago, people regularly struggle to access basic necessities such as groceries and toiletries, often needing to travel for hours to buy everyday goods. Meanwhile, producers experience difficulties reaching far-flung customers, due to both an absence of physical and digital infrastructure.

Dagangan (<https://www.dagangan.com/>) was founded in 2019 by a group of friends with first-hand knowledge of the struggles of rural Indonesians to achieve dreams and raise living standards, seeking to empower them "not only to survive but thrive." Led by Ryan Manafe and Wilson Yanaprasetya, they created a rural commerce platform to bring local communities access to affordable, quality goods, with no additional logistical costs, in under 24 hours. Inspired by the UN Sustainable Development Goals (UN SDGs), Dagangan aims to unleash the economic potential of Indonesia's agricultural heartland through tech-enabled community ecosystems.

Dagangan tapped a keen unmet demand that enabled it to grow rapidly across the main Indonesian island of Java. Success brought with it the challenge of system reliability and availability in a multi-cloud environment that crashed at times of peak demand, as well as escalating costs provisioning virtual machines (VMs).

Working with partner Devoteam

(<https://www.devoteam.com/>), the startup embarked upon an ambitious migration to Google Cloud to enable a no-limits journey to success. It chose Google Cloud (<https://cloud.google.com/>) thanks to the launch of a new Jakarta data center, key to its mission of system reliability, and the "exciting possibilities" enabled by a seamless end-to-end Google Cloud architecture based on Google Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine>) (GKE) and BigQuery (<https://cloud.google.com/bigquery>).

Dagangan also enacted a Google Workspace (<https://workspace.google.com/>) adoption to optimize the creative collaboration needed to rapidly develop the platform in a secure and seamless fashion.

Since Google Cloud and Google Workspace adoption, Dagangan's growth trajectory has been impressive. The startup has rapidly expanded to 50 hubs across Java, reaching more than 20,000 villages, while enabling nearly half of the family owned-businesses in its network to increase their revenue by 2x.

"We're impatient to grow our solution across Indonesia to improve the lives of rural Indonesians,"

says Wilson Yanaprasetya, President and Co-founder of Dagangan. "We couldn't scale fast enough on our original architecture. Now with a combination of Google Kubernetes Engine and BigQuery, alongside Google Workspace productivity tools, we're confident we can continue our rapid growth journey to every corner of Indonesia, and beyond, with confidence and peace of mind."

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Empowering remote Indonesian communities with BigQuery and GKE transformation

The Dagangan innovation model is built upon complex ecosystems of local community actors, brought together through crowdsourcing, who enable deliveries of goods collected at central hubs that fan out in spoke-like fashion to points across each Dagangan delivery zone.

At the heart of the system is rural women, whom Dagangan seeks to empower in accordance with the UN SDG of gender equality. 90% of Dagangan users are women, often homemakers who have never held a job. They place orders from village shops or family homes, for those who lack devices to order directly from the app, empowering themselves as well as the community.

Community leaders are also a vibrant part of the ecosystem. These local community leaders often double as Dagangan drivers, using spare vehicles and local knowledge for timely deliveries to hard-to-find drop off points. In countless villages, meanwhile, neighbors regularly come together to aggregate the entire village demand so an agent can place an order in bulk.



It's a complex web of community interaction that requires data-driven insight on inventory and routing, as well as agile scalability, to navigate a shifting landscape of logistical complexities. By moving from its legacy VM-based architecture to a combination of GKE and BigQuery, Dagangan is able to optimize its warehousing, logistics, and crowdsourcing requirements with ease and confidence.

Originally, for example, the Dagangan team needed to manually install Kubernetes clusters in VMs to drive the scale needed for its network, a cumbersome, expensive and time-consuming

process. By switching to GKE as its fully-managed microservices solution, Dagangan gained the nimble auto scaling it needs to anticipate and meet any spike due to sales promotions or other factors. It also freed up the DevOps team to focus on creative solution building instead of mundane provisioning tasks.

Meanwhile, BigQuery is the enabler of data-driven workflows as Dagangan ingests a continuous stream of data on user transactions, drop-off locations, inventory, and routing patterns. This gives Dagangan constantly improving ML-enabled intelligence on how to increase the speed and efficiency of deliveries. The team also deploys BigQuery to analyze platform access logs for snapshots of drop-off rates on app pages. This enables targeted platform improvements that boost user retention.

The end-to-end Google Cloud architecture, which also includes Cloud SQL (<https://cloud.google.com/sql>) as Dagangan's fully-managed database, has unlocked important results. The platform has grown its daily active users from 3,000 to 5,000 in just one year since migrating to Google Cloud. The reliability of the platform has also increased, with the percentage of users enjoying a crash-free experience rising from 92% to more than 99%.

"Most importantly, Google Cloud brings us the unlimited auto scaling we need to drive an open-ended expansion narrative," says Andhika Estrada, CTO, Dagangan. "Since migration, we've enjoyed a rapid traffic growth from 152 million user requests

to more than 200 million, all while seeing platform reliability improve significantly. That's a game-changing difference."

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Enabling agile and creative collaboration with Google Workspace synergies

In order to build the future of rural commerce across Indonesia and beyond, Dagangan relies on Google Workspace to give it the collaboration capabilities needed to innovate with nimble efficiency, in a secure-by-design environment, wherever developers are based.

By using a combination of [Docs](#)

(<https://workspace.google.com/products/docs/>), [Drive](#)

(<https://workspace.google.com/products/drive/>), [Sheets](#)

(<https://workspace.google.com/products/sheets/>) and

[Google Meet](#)

(<https://workspace.google.com/products/meet/>),

Dagangan is able to divide its workforce into any number of fluid project teams, with each member given finely-tuned levels of access and permissions.

In such an agile, cloud-based work environment, Dagangan is able to assemble teams in real time according to the needs of the moment, to drive productivity and collaborative synergies. In fact, the organization has a unique initiative called, 'Working from wherever you're the happiest,' which, in the most literal sense allows employees to work from

anywhere they feel their best selves. Google Workspace has been a key enabler for this initiative, as it allows employees to maintain a high level of productivity and a strong collaborative culture.

"As much as scalability and data analytics, creative team synergies are required for us to achieve our mission of social impact," says Yanaprasetya.

"Google Workspace tools such as Docs, Sheets and Meet give us the secure, team-building capabilities we need to fast invent the future of our platform for the betterment of society."



A future of Southeast Asian expansion with Google AI and Google Maps solutions

Looking into the future, Dagangan has ambitions to drive its social impact across Southeast Asia. The platform plans to grow from its current market of Java throughout Indonesia's sprawling archipelago, then to countries such as the Philippines, Vietnam

and Thailand "which have similar social structures to ours."

To drive this strategy, Dagangan sees adoption of Google Cloud machine learning and AI capabilities as a key factor in enabling both the security protocols and data-driven insights it needs to power its expansion. Estrada envisions deploying solutions such as BigQuery ML

(<https://cloud.google.com/bigquery-ml/docs>) to support delivery efficiencies through ML modeling of fastest routing and optimal drop off spots. He also

envisions usage of Vision AI

(<https://cloud.google.com/vision>) for know-your-customer (KYC) purposes, such as to uncover fraudulent users by matching photo documents with national identification cards.

"We want to go deeper and deeper into rural communities, growing 10 times bigger in the next three years," says Yanaprasetya. "We're confident that only Google Cloud enables us to achieve that scale. Now adoption of Google Cloud AI solutions can help fuel a journey of dynamic empowerment across Southeast Asia."

Yanaprasetya also sees a more proactive and creative usage of Google Maps Platform

(<https://mapsplatform.google.com/>) APIs to bring superior routing and delivery abilities, as Dagangan moves more deeply into remote Indonesian areas.

"Google Maps is out in front in terms of mapping far-reaching places that nobody else has mapped. So it will be a key enabler as we expand," says

Yanaprasetya. "Everything for us is about speed of execution. The faster we execute, the faster we can improve lives and people's income. Google Cloud AI solutions can combine with Google Maps to propel this journey of growth and empowerment."

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